Computer Networks

Overview of the Internet
Protocol layering (Book 1 Ch 1)

Application Layer

- Application-Layer Paradigms
- Client-server applications
    - World Wide Web and HTTP
    - FTP
    - Electronic Mail
    - DNS
- Peer-to-peer paradigm
    - P2P Networks
- Case study: BitTorrent (Book 1 Ch 2)

Application Layer

1. **Introduction**
   - Providing Services
   - Application-Layer Paradigms
2. **Standard Client-Server Applications**
   - World Wide Web and HTTP
   - FTP
   - Electronic Mail
   - Domain Name System (DNS)
3. **Peer-to-Peer Paradigm**
   - P2P Networks

Domain Name System (DNS)

Definition

The **Domain Name System (DNS)** is a distributed client–server application that translates human-readable domain names into IP addresses.

Why DNS is Needed

- TCP/IP identifies hosts using IP addresses.
- Humans prefer meaningful names instead of numbers.
- DNS acts as the Internet's directory service.
- Similar to a telephone directory that maps names to phone numbers.

Problem with Centralized Directory

- Huge Internet cannot be managed by one central directory.
- Heavy traffic overloads a single server.
- Failure of one server stops name resolution.

DNS Client–Server Architecture

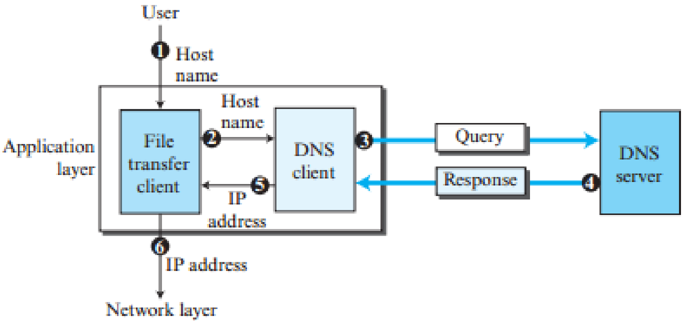
DNS follows the Client–Server model.

1. User enters the server name (e.g., `afilesource.com`).
2. Application passes the name to the DNS Resolver.
3. Resolver sends a query to the local DNS Server.
4. DNS Server returns the corresponding IP address.
5. Resolver returns the IP address to the application.
6. Application communicates using the obtained IP address.

Key Point

DNS resolution always occurs before communication with the destination host.

Figure 2.35 *Purpose of DNS*



Name Space

Definition

A **Name Space** is a collection of unique names that are mapped to unique IP addresses.

Flat Name Space

- Single-level naming
- No hierarchy
- Centrally managed
- Difficult to scale
- Not suitable for the Internet

Hierarchical Name Space

- Multi-level naming
- Organized tree structure
- Distributed management
- Highly scalable
- Used by DNS

Exam Point

DNS uses a **Hierarchical Name Space** because it supports distributed administration and scalability.

Domain Name Space

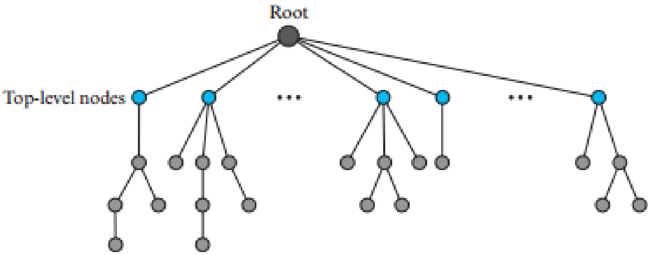
Domain Name Space

DNS organizes names using a hierarchical **inverted tree structure**.

Characteristics

- Root is at Level 0.
- Maximum depth is 128 levels.
- Every node has a unique label.
- Children of the same parent cannot have identical labels.
- Domain names are formed by joining labels with dots (.).

Figure 2.36 *Domain name space*



Labels, Domain Names, FQDN and PQDN

Label

- String with a maximum of 63 characters.
- Root has a null (empty) label.
- Labels under the same parent must be unique.

Domain Name

- Sequence of labels separated by dots.
- Read from node toward the root.

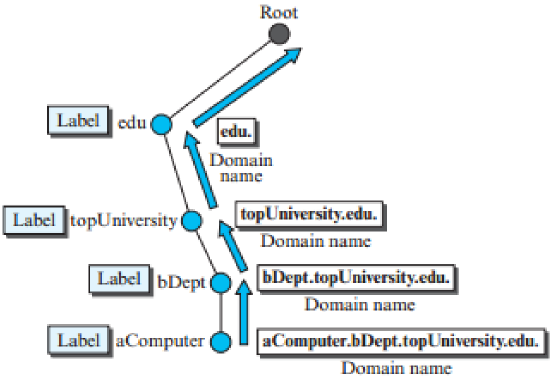
FQDN (Fully Qualified Domain Name)

- Complete domain name ending with a dot (.).
- Example: `www.example.com.`

PQDN (Partially Qualified Domain Name)

- Incomplete domain name.
- Resolver appends the missing suffix.

Figure 2.37 *Domain names and labels*



Domain and Distribution of Name Space

Domain

A **Domain** is a subtree of the DNS hierarchy.

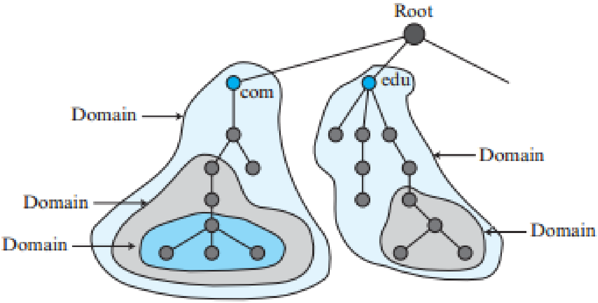
Features

- Named after the top node.
- May contain several subdomains.
- Authority can be delegated.
- Makes DNS scalable.

Distribution of Name Space

- Entire DNS database cannot be stored on one server.
- Information is distributed among many DNS servers.
- Each server is responsible for its assigned domain.

Figure 2.38 *Domains*



Hierarchy of Name Servers

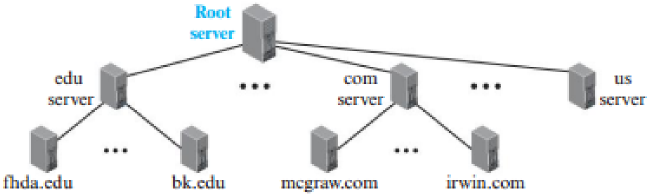
DNS servers are organized hierarchically.

- Root Server
- Top-Level Domain (TLD) Server
- Authoritative Name Server
- Local DNS Server

Advantages

- Distributed management
- Faster lookup
- Scalability
- Fault tolerance

Figure 2.39 *Hierarchy of name servers*



Zone, Root Server, Primary and Secondary Servers

Zone

- Portion of DNS namespace managed by one server.
- If no delegation:

$$\boxed{\text{Domain} = \text{Zone}}$$

- With delegation, a zone becomes part of a domain.

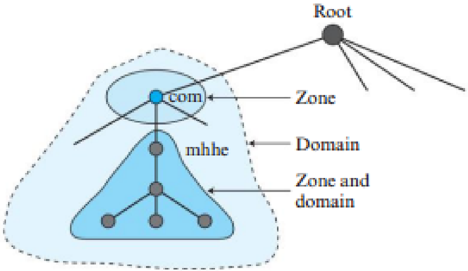
Root Server

- Responsible for the root of the DNS tree.
- Delegates requests to Top-Level Domain (TLD) servers.

Primary vs Secondary Server

- Primary: Creates and updates zone files.
- Secondary: Receives copies of zone files for redundancy.
- Both are authoritative for their zones.

Figure 2.40 *Zone*



DNS in the Internet

Originally, the DNS namespace was divided into:

1. Generic Domains
2. Country Domains
3. Inverse Domain (Deprecated - RFC 3425)

Generic Domain Labels

Label	**Description**
com	Commercial organizations
edu	Educational institutions
gov	Government organizations
org	Non-profit organizations
net	Network support centers
mil	Military organizations
biz	Businesses
info	Information providers
name	Personal names
pro	Professional organizations
coop	Cooperative organizations
aero	Airlines and aerospace

Country Domains

Country Domains

Country domains use two-letter country codes based on ISO standards.

Examples

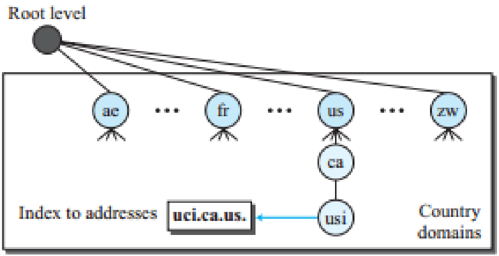
- in – India
- us – United States
- uk – United Kingdom
- jp – Japan
- au – Australia

Example:

`uci.ca.us`

University of California, Irvine, California, USA.

Figure 2.42 *Country domains*



Name Resolution

Definition

Mapping a domain name to an IP address is called **Name Resolution**.

Resolution Steps

1. Application sends the domain name.
2. Resolver sends query to Local DNS Server.
3. Local DNS Server searches its database.
4. If unavailable, it contacts other DNS servers.
5. Destination IP address is returned.
6. Communication starts using the obtained IP address.

Exam Point

The DNS Resolver is the client responsible for initiating name resolution.

Recursive Resolution

Definition

Each DNS server is responsible for obtaining the final answer before replying.

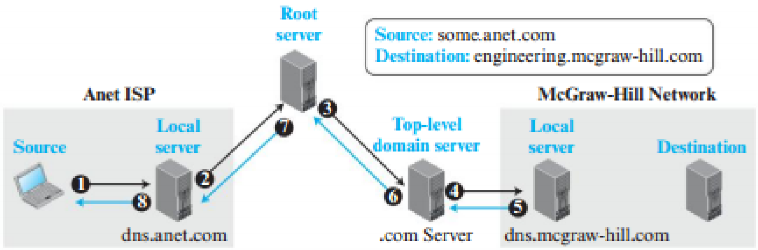
Sequence

1. Resolver
2. Local DNS Server
3. Root Server
4. Top-Level Domain Server
5. Authoritative DNS Server
6. Final IP address returned through the same path

Advantages

- Simple for the client
- Client receives the final answer

Figure 2.43 *Recursive resolution*



Iterative Resolution

Definition

Each DNS server returns the address of the next DNS server instead of the final answer.

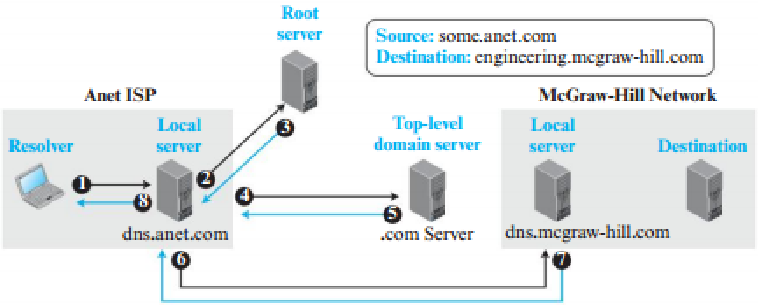
Sequence

1. Resolver queries Local DNS Server.
2. Local DNS Server queries Root Server.
3. Root Server returns TLD Server address.
4. TLD Server returns Authoritative Server address.
5. Local DNS Server queries the Authoritative Server.
6. Final IP address is returned.

Recursive vs Iterative

- Recursive: Returns final answer.
- Iterative: Returns next server address.

Figure 2.44 *Iterative resolution*



DNS Caching and Time To Live (TTL)

DNS Caching

Caching stores recently resolved DNS records temporarily to speed up future queries.

Advantages

- Faster name resolution
- Reduces network traffic
- Decreases load on higher-level DNS servers
- Improves overall DNS performance

Time To Live (TTL)

- Specifies how long a cached record remains valid.
- After TTL expires, the cached record is removed.
- A new query is sent to the authoritative DNS server.

Exam Point

TTL prevents outdated DNS information from being used.

Resource Records (RR)

Definition

DNS stores information as **Resource Records (RR)**.

Resource Record Format

$$(\text{Domain Name}, \text{Type}, \text{Class}, \text{TTL}, \text{Value})$$

Common RR Types

Type	**Purpose**
A	IPv4 Address
AAAA	IPv6 Address
NS	Authoritative Name Server
CNAME	Alias Name
MX	Mail Server
SOA	Start of Authority

Exam Point

A $\rightarrow$ IPv4 AAAA $\rightarrow$ IPv6 MX $\rightarrow$ Mail Server

DNS Messages and Encapsulation

DNS Messages

- Query Message
- Response Message

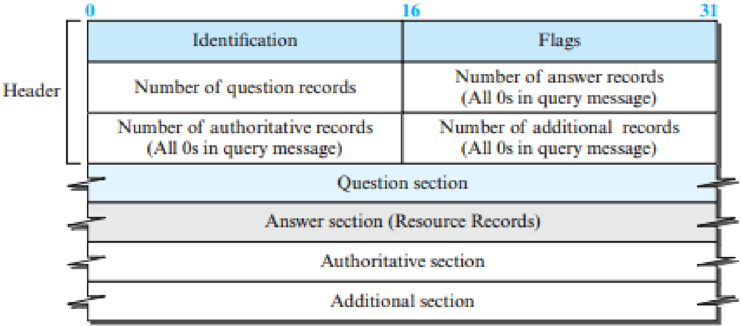
Each message contains

- Header
- Question Section
- Answer Section
- Authority Section
- Additional Information Section

Encapsulation

- Uses UDP or TCP.
- Well-known Port Number: **53**.

Figure 2.45 *DNS message*



Note:
The query message contains only the question section.
The response message includes the question section,
the answer section, and possibly two other sections.

DNS Operation and Dynamic DNS

UDP vs TCP

- UDP used for responses smaller than 512 bytes.
- TCP used for Zone Transfer and large responses.
- If UDP response exceeds 512 bytes, the server sets the TC (Truncated) bit and the client repeats the request using TCP.

Registrars

- Accredited by ICANN.
- Register unique domain names.

Dynamic DNS (DDNS)

- Automatically updates DNS records.
- Frequently works with DHCP.
- Supports Active and Passive notifications.
- Uses Zone Transfer to update secondary servers.

DNS Security and Summary

DNS Security Threats

- Eavesdropping on DNS queries
- DNS Spoofing / Bogus Responses
- Denial of Service (DoS) attacks

DNSSEC

- Provides Message Origin Authentication
- Provides Message Integrity
- Uses Digital Signatures
- Does NOT provide Confidentiality

Key Takeaways

- DNS Architecture
- Name Space
- DNS Servers
- Resolution
- Caching
- Resource Records
- DNS Messages
- UDP/TCP
- DDNS
- DNSSEC